

BS – 101 Engineering Physics

Lecture – 9

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Energy Storage in Capacitors

- Many of the most important applications of capacitors depend on their ability to store energy.
- The electric potential energy stored in a charged capacitor is just equal to the amount of work required to charge it—that is, to separate opposite charges and place them on different conductors.
- When the capacitor is discharged, this stored energy is recovered as work done by electrical forces.



Energy Storage in Capacitors (Contd.)

- We can calculate the potential energy U of a charged capacitor by calculating the work W required to charge it.
- Suppose that when we are done charging the capacitor, the final charge is Q and the final potential difference is V . These quantities are related by
- $V = \frac{Q}{C}$

Energy Storage in Capacitors (Contd.)

- Let q and v be the charge and potential difference, respectively, at an intermediate stage during the charging process; then;

- $v = \frac{q}{C}$

- At this stage the work dW required to transfer an additional element of charge dq is;

- $dW = v dq = \frac{q dq}{C}$

Energy Storage in Capacitors (Contd.)

- The total work W needed to increase the capacitor charge q from zero to Q is;
- $W = \int_0^W dW = \frac{1}{C} \int_0^Q q dq = \frac{Q^2}{2C}$
- This is also the total work done by the electric field on the charge when the capacitor discharges. Then q *decreases* from an initial value Q to zero as the elements of charge dq “fall” through potential differences v that vary from V down to zero.

Energy Storage in Capacitors (Contd.)

- If we define the potential energy of an *uncharged* capacitor to be zero, then W in previous Eq. is equal to the potential energy U of the charged capacitor. The final stored charge is $Q = CV$, so we can express U (which is equal to W) as;

- $$U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV$$

Energy Density

- We can charge a capacitor by moving electrons directly from one plate to another. This requires doing work against the electric field between the plates. Thus we can think of the energy as being stored *in the field* in the region between the plates. To see this, let's find the energy *per unit volume* in the space between the plates of a parallel-plate capacitor with plate area A and separation d . We call this the **energy density**, denoted by u . From previous Eq. the total stored potential energy is $\frac{1}{2} CV^2$ and the volume between the plates is Ad ; hence;

- $u = \text{Energy Density} = \frac{\frac{1}{2} CV^2}{Ad}$



Energy Density

- From previous Eq. the capacitance C is given by $C = \frac{\epsilon_0 A}{d}$. The potential difference V is related to the electric-field magnitude E by $V = Ed$. If we use these
- expressions in previous Eq., the geometric factors A and d cancel, and we find;
- $u = \frac{1}{2} \epsilon_0 E^2$



Transferring Charge and Energy Between Capacitors

- We connect a capacitor $C_1 = 8.0 \mu F$ to a power supply, charge it to a potential difference $V_0 = 120 V$, and disconnect the power supply. Switch S is open. (a) What is the charge Q_0 on C_1 ? (b) What is the energy stored in C_1 ? (c) Parallel Capacitor $C_2 = 4.0 \mu F$ is initially uncharged. We close switch S . After charge no longer flows, what is the potential difference across each capacitor, and what is the charge on each capacitor? (d) What is the final energy of the system?



Electric Field Energy

- (a) What is the magnitude of the electric field required to store **1.00 J** of electric potential energy in a volume of **1.00 m³** in vacuum? (b) If the field magnitude is 10 times larger than that, how much energy is stored per cubic meter?



Dielectrics

- Most capacitors have a nonconducting material, or **dielectric**, between their conducting plates. A common type of capacitor uses long strips of metal foil for the plates, separated by strips of plastic sheet such as Mylar. A sandwich of these materials is rolled up, forming a unit that can provide a capacitance of several microfarads in a compact package.



Dielectrics (Contd.)

- Placing a solid dielectric between the plates of a capacitor serves three functions.
- First, it solves the mechanical problem of maintaining two large metal sheets at a very small separation without actual contact.



Dielectrics (Contd.)

- Second, using a dielectric increases the maximum possible potential difference between the capacitor plates. Any insulating material, when subjected to a sufficiently large electric field, experiences a partial ionization that permits conduction through it. This phenomenon is called **dielectric breakdown**. Many dielectric materials can tolerate stronger electric fields without breakdown than can air. Thus using a dielectric allows a capacitor to sustain a higher potential difference V and so store greater amounts of charge and energy.



Dielectrics (Contd.)

- Third, the capacitance of a capacitor of given dimensions is *greater* when there is a dielectric material between the plates than when there is vacuum.

Dielectrics (Contd.)

- The original capacitance C_0 is given by $C_0 = \frac{Q}{V_0}$, and the capacitance C with the dielectric present is $C = \frac{Q}{V}$. The charge Q is the same in both cases, and V is less than V_0 , so we conclude that the capacitance C with the dielectric present is *greater* than C_0 . When the space between plates is completely filled by the dielectric, the ratio of C to C_0 (equal to the ratio of V_0 to V) is called the **dielectric constant** of the material, K :
- $K = \frac{C}{C_0}$ (definition of dielectric constant)

Dielectrics (Contd.)

- When the charge is constant, $Q = C_0 V_0 = CV$ and $\frac{C}{C_0} = \frac{V_0}{V}$. In this case,
- $V = \frac{V_0}{K}$ (when Q is constant)
- With the dielectric present, the potential difference for a given charge Q is *reduced* by a factor K .
- The dielectric constant K is a pure number. Because C is always greater than C_0 , K is always greater than unity.



Dielectrics (Contd.)

Values of Dielectric Constant K at 20°C

Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas [®]	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar	3.1	Strontium titanate	310



Dielectrics (Contd.)

- Dielectric Permittivity
- $\epsilon = K\epsilon_0$



A Capacitor With and Without a Dielectric

- Suppose the parallel plates capacitor and each plate have an area of **2000 cm^2 ($2.00 \times 10^{-1} \text{ m}^2$)** and are **1.00 cm ($1.00 \times 10^{-2} \text{ m}$)** apart. We connect the capacitor to a power supply, charge it to a potential difference **$V_0 = 3.00 \text{ kV}$** , and disconnect the power supply. We then insert a sheet of insulating plastic material between the plates, completely filling the space between them. We find that the potential difference decreases to **1.00 kV** while the charge on each capacitor plate remains constant. Find (a) the original capacitance **C_0** ; (b) the magnitude of charge **Q** on each plate; (c) the capacitance **C** after the dielectric is inserted; (d) the dielectric constant **K** of the dielectric; (e) the permittivity **ϵ** of the dielectric; (f) the original electric field **E_0** between the plates; and (g) the electric field **E** after the dielectric is inserted.



Energy Storage With and Without a Dielectric

- Find the energy stored in the electric field of the capacitor in previous example and the energy density, both before and after the dielectric sheet is inserted.